

Department of Computer Science and Engineering

Capstone Project Evaluation and Guidelines

Academic Year	2026-27		Date	27-05-2025
Course Name	Machine Learning	Course Code: 23CSE301	L-T-P-C: 3-0-2-4	
Faculty Member(s)	Dr. S. S. Chakravarthi, Ms. K V Meenatchi			

I. Introduction

The Capstone Project provides an excellent opportunity for the students to engage in “high-level work” focusing on an area of specialization within the Machine Learning and allied domains. These are expected to be inquiry and practice-centred, and would draw upon areas of interest to the student. All Capstones projects aim to bridge theory and practice and are aimed to have an impact on the professional life of students whether they work in classrooms, studios or community spaces.

II. Capstone Project Problem Selection

Capstone projects often take their inspiration from projects, papers, and experiences related to course work (in this case Machine Learning). However, to ascertain students’ abilities for independent work and their capacity for self-directed inquiry, capstone projects must demonstrate in what ways individual graduate students have researched, developed, and extended, or applied the ideas and strategies under investigation.

A repository of research papers (in Machine Learning and allied application domains) will be made available to the students. For their capstone project, Students shall identify **ONE** Research Papers (henceforth called as Base papers) from the repository and communicate to the faculty through the MS Forms. (link will be provided by the faculty). The problem selection is **purely on First Come First Serve basis** and all the discrepancies shall be solved in consultation with the faculty only.

Research Papers Repository

[AY2026-27 - OneDrive](#)

(<https://amritacampuschennai-my.sharepoint.com/my?id=%2Fpersonal%2Fss%5Fchakravarthi%5Fch%5Famrita%5Fedu%2FDocuments%2FML%20Research%20Papers%20Repository%2FAY2026%2D27&ga=1>)

Communication of the Selected Problem

<https://docs.google.com/forms/d/e/1FAIpQLSf95vm6LavoSF-0JCqpgyFTJ6t4opQITPwNWCGmnWxWAS5t0A/viewform?usp=header>

III. Evaluation Grading

The Capstone project evaluation is in line with the Machine Course Evaluation and grading policy as under:

Machine Learning Evaluation and Grading					
Internal (70)				External (30)	Total
Components		Marks	Total Marks	End Semester = 30	Internal + External = 100
Mid term		20	20		
Continuous Assessment (Theory)	Assignments (3)	10	50		
Continuous Assessment (Lab): Capstone Project	Lab Activities (6)	05			
	Foundation and Intermediate Level Certification in Machine Learning by Infosys Springboard	10			
	Phase – 1: Problem Statement Definition and Interpretation	05			
	Phase – 2: Features Selection & Learning Technique Identification	10			

	Phase – 3: End Sem Capstone Evaluation	10			
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CO Attainment

		Marks	CO1	CO2	CO3	CO4	CO5
Internal	Midterm	20	✓		✓		✓
	Assignments (3)	10	✓	✓	✓	✓	✓
	Lab Activities (6)	05	✓	✓			
	Industry Certification	10	✓			✓	✓
	Phase – 1	05	✓	✓	✓		
	Phase – 2	10		✓	✓	✓	✓
	Phase – 3	10	✓	✓	✓	✓	✓
External	End Sem	30	✓	✓	✓	✓	✓

Assessment Rubrics

The Capstone Project components are assessed based on the following Rubrics:

1. Assignment Evaluation Rubrics

Assignment: (50 Marks)			
Evaluation Component	Excellent (3)	Good (2)	Not Impressive (1)
Assignment Content & Presentation (30 Marks)	The student incorporated all the suggestions given during the Lab Evaluation (30x1=30 Marks)	The student incorporated 50% of the suggestions given during the Lab Evaluation (30x0.7=21 Marks)	The student incorporated 20% to 30% of the suggestions given during the Lab Evaluation (30x0.5=15 Marks)
Plagiarism & AI Tools Usage (20 Marks)	The similarity index of the assignment is less than 10% with 30% AI Tools usage (20x1=20 Marks)	The similarity index of the assignment is between 10% to 30% with 30% to 45% AI Tools usage (20x0.7=14 Marks)	The similarity index of the assignment is More than 30% with higher than 45% AI Tools usage (20x0.5=10 Marks)

2. Phase – 1: Problem Statement Definition and Interpretation

Phase – 1: Problem Statement Definition and Interpretation (100 Marks)			
Evaluation Component	Excellent (1)	Good (0.6)	Not Impressive (0.4)
Selection of the Problem (30 Marks)	The problem is real time and has scope to work (30x1=30 Marks)	The problem is not a real-time but has scope to work (30x0.6=18 Marks)	The problem maybe a real-time but has limited scope to work (30x0.4=12 Marks)
Understanding of the Problem (30 Marks)	The student has in-depth understanding of the problem and has clarity on how to proceed with the "todo" work (30x1=30 Marks)	The student has understanding of the problem, but need more clarity on how to proceed with the "todo" work (30x0.6=18 Marks)	The student has minimal understanding of the problem and absolutely no idea on how to proceed with the "todo" work (30x0.4=12 Marks)
Identifying the datasets used and Interpretation of the Solution given on the dataset (30 Marks)	The student has clearly identified the dataset used and the ML technique proposed in the base paper (30x1=30 Marks)	The student has identified the dataset used and the ML technique proposed in the base paper, but lacks deep understanding & clarity (30x0.6=18 Marks)	The student has not identified the dataset used and no clarity on the ML technique proposed in the base paper (30x0.4=12 Marks)
Presentation & Interaction (10 Marks)	Student delivered his content clearly orally & verbally (10x1=10 Marks)	Student delivered his content orally & verbally, but with minimal errors (10x0.6=06 Marks)	Student delivered his content orally & verbally, but with blunders (10x0.4=04 Marks)

3. Phase – 2: Features Selection and Learning algorithm / technique identification

Phase – 2: Features Selection and Learning algorithm / technique identification. (100 Marks)			
Evaluation Component	Excellent (1)	Good (0.6)	Not Impressive (0.4)
Feature selection (30 Marks)	The student has clearly identified the features to be used for the identified problem (30x1=30 Marks)	The student identified the features but not suitable to use for the identified problem (30x0.6=18 Marks)	The student has not identified the features for the identified problem (30x0.4=12 Marks)
Identification of existing Algorithm through Literature Survey (30 Marks)	The student has clearly identified the existing ML techniques proposed till date for the identified problem, through an extensive literature survey (30x1=30 Marks)	The student has identified the existing ML techniques proposed for the identified problem, but through moderate level of literature survey. (30x0.6=18 Marks)	The student has limited knowledge on existing ML techniques proposed for the identified problem, as his literature survey was very narrow (30x0.4=12 Marks)
Identification of Learning algorithm / technique / method (30 Marks)	The student has identified the learning technique / method suitable for the proposed work with appropriate justification (30x1=30 Marks)	The student has identified the learning technique / method suitable for the proposed work with non-convincing justification (30x0.6=18 Marks)	The student has not identified the learning technique / method and not inclined towards solving the problem (30x0.4=12 Marks)
Presentation & Interaction (10 Marks)	Student delivered his content clearly orally & verbally (10x1=10 Marks)	Student delivered his content orally & verbally, but with minimal errors (10x0.6=06 Marks)	Student delivered his content orally & verbally, but with blunders (10x0.4=04 Marks)

4. Phase – 3: End Sem Capstone Evaluation

Capstone Project (150 Marks)			
Evaluation Component	Excellent (3)	Good (2)	Not Impressive (1)
Problem Identification and Literature survey (40 Marks)	The student has in-depth understanding of the problem and has highlighted gaps in existing literature and areas for future research. (40x1=40 Marks)	The student understanding of the problem, but lacks in highlighting the gaps in existing literature and areas for future research. (40x0.6=24 Marks)	The student has minimal understanding of the problem and has no idea of identifying research gaps in existing literature. (40x0.4=16 Marks)
Handling datasets and Data Preprocessing (40 Marks)	The student has Clearly identified data sources, addressed the inconsistencies in data and Justified for the extraction of specific features. (40x1=40 Marks)	The student has identified data sources but lacks in addressing the inconsistencies in data and Justification for the extraction of specific features. (40x0.6=24 Marks)	The student has identified data sources, but could not address the data inconsistencies and inappropriate justification on features extraction. (40x0.4=16 Marks)
Algorithmic Design & Approach (40 Marks)	The student has commendable knowledge on the proposed algorithm and produces convincing results for various test cases (40x1=40 Marks)	The student just has knowledge on the proposed algorithm, with limited results or no improvement on the benchmarked results. (40x0.6=24 Marks)	The student has very minimal knowledge on the proposed algorithm and no convincing results for various test cases. (40x0.4=16 Marks)
Presentation & Interaction (30 Mark)	Student delivered his content clearly orally & verbally (30x1=30 Marks)	Student delivered his content orally & verbally, but with minimal errors (30x0.6=18 Marks)	Student delivered his content orally & verbally, but with blunders (30x0.4=12 Marks)

IV. Timelines

The Capstone project reviews would be conducted in the Lab continuously, and the due dates for each component is as under:

Evaluation Component	Due Date
Assignments (3)	One Week after their Review(s)
Phase – 1: Problem Statement Definition and Interpretation	Before 30-July
Phase – 2: Features Selection & Learning Technique Identification	Before 27-Aug
Phase – 3: End Sem / Final Evaluation	Before 15-Oct

V. General Guidelines

The following are a few general guidelines to the students which helps them to complete the Capstone Project effectively.

- Timelines are firm. No changes in timelines are entertained without prior approval of the faculty.**
- The students should identify and select a specific problem going through the repository, based on their interest, knowledge and expertise.
- The students are expected to and should focus on:
 - Identification of Datasets / Collection of data relevant to the identified problem.
 - Use Data Analysis tools and gain 'problem-oriented' interpretation on the data.
 - Note down the findings, observations, suggestions and recommendations during the reviews for improving the approach / work.
 - Learn new techniques / concepts that come in their way to complete the project.
- Each capstone project must show "depth and breadth of scholarly understanding", and "the application of this understanding to practice".
- The students should demonstrate the ability to gather materials, review current literature, and to examine sufficient background material to acknowledge the original work.
- The work is expected to emphasize good writing, formatting, attention to detail and mistake-free work. Work submitted should be free of typos, incorrect grammar and other avoidable errors. All work should be self-edited and proofread before submission.



Faculty



Course Mentor

Program Chair

Chairperson